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OmniFood8K: Single-Image Nutrition Estimation via Hierarchical Frequency-Aligned Fusion

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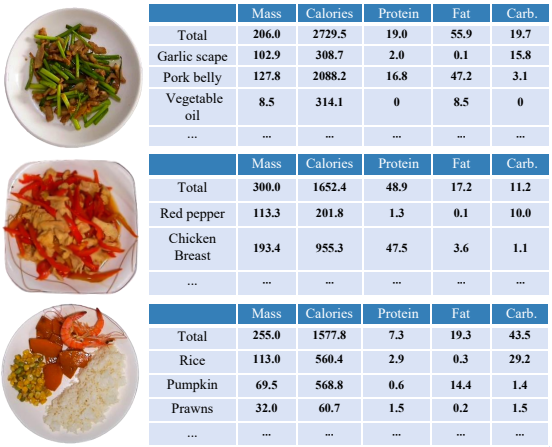
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Abstract

Accurate estimation of food nutrition plays a vital role in promoting healthy dietary habits and personalized diet management. Most existing food datasets primarily focus on Western cuisines and lack sufficient coverage of Chinese dishes, which restricts accurate nutritional estimation for Chinese meals. Moreover, many state-of-the-art nutrition prediction methods rely on depth sensors, restricting their applicability in daily scenarios. To address these limitations, we introduce OmniFood8K, a comprehensive multimodal dataset comprising 8,036 food samples, each with detailed nutritional annotations and multi-view images. In addition, to enhance models' capability in nutritional prediction, we construct NutritionSynth-115K, a large-scale synthetic dataset that introduces compositional variations while preserving precise nutritional labels. Moreover, we propose an end-to-end framework for nutritional prediction from a single RGB image. First, we predict a depth map from a single RGB image and design the Scale-Shift Residual Adapter (SSRA) to refine it for global scale consistency and local structural preservation. Second, we propose the Frequency-Aligned Fusion Module (FAFM) to hierarchically align and fuse RGB and depth features in the frequency domain. Finally, we design a Mask-based Prediction Head (MPH) to emphasize key ingredient regions via dynamic channel selection for more accurate prediction. Extensive experiments on multiple datasets demonstrate the superiority of our method over existing approaches. Project homepage: https://yudongjian.github.io/OmniFood8K-food/

1. Introduction

With the improvement of living standards and the diversification of dietary habits, nutritional assessment of food has become a key research focus in both public health and personal health management [28, 38, 57]. Accurate estimation of food energy and nutrient intake plays a vital role in preventing chronic diseases such as obesity, diabetes, and cardiovascular disorders [22, 45]. However, traditional nutritional assessment methods [35, 48, 59] often rely on manual recording or expert estimation, which are not only time-consuming and labor-intensive but also prone to subjective bias [20, 32]. This makes accurate and consistent evaluation challenging, particularly in real-world daily dietary applications [14, 37].



	Mass	Calories	Protein	Fat	Carb.
Total	206.0	2729.5	19.0	55.9	19.7
Garlic scape	102.9	308.7	2.0	0.1	15.8
Pork belly	127.8	2088.2	16.8	47.2	3.1
Vegetable oil	8.5	314.1	0	8.5	0
...	...	...	...	...	...

	Mass	Calories	Protein	Fat	Carb.
Total	300.0	1652.4	48.9	17.2	11.2
Red pepper	113.3	201.8	1.3	0.1	10.0
Chicken Breast	193.4	955.3	47.5	3.6	1.1
...	...	...	...	...	...

	Mass	Calories	Protein	Fat	Carb.
Total	255.0	1577.8	7.3	19.3	43.5
Rice	113.0	560.4	2.9	0.3	29.2
Pumpkin	69.5	568.8	0.6	14.4	1.4
Prawns	32.0	60.7	1.5	0.2	1.5
...	...	...	...	...	...

Figure 1. Representative examples from the OmniFood8K dataset.

Recent advances in computer vision and multimodal learning have significantly advanced the development of automated food nutrition assessment [10, 11, 42–44, 69]. Deep learning techniques have shown great potential in automatically recognizing, segmenting, analyzing food images to estimate food mass, volume, and nutritional composition [16, 47, 58]. Moreover, leveraging multimodal features such as textual descriptions and sensor data can further enhance estimation accuracy and robustness, facilitating personalized nutrition monitoring and healthcare sys-

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tems [8, 53].

Despite these advances, current methods still encounter two fundamental limitations that constrain their practical usability. **(1) Data Limitation.** Existing datasets are heavily biased toward Western cuisines, with limited representation of Chinese food. The inherent diversity, complex ingredient composition, and non-standardized preparation of Chinese food make reliable annotation and quantitative nutritional analysis particularly challenging. **(2) Algorithmic Limitation.** Many advanced approaches rely on depth maps captured by depth cameras for accurate nutrition estimation. Nevertheless, in most real-world contexts, food photographs are taken with conventional RGB cameras rather than depth sensors, thereby restricting the practical adoption of these methods.

To address these two limitations, this work makes two key contributions. First, we introduce OmniFood8K, a comprehensive multimodal food nutrition dataset designed for real-world scenarios, as shown in Figure 1. The dataset covers the entire food preparation process, including raw ingredient images with mass information, structured recipe descriptions, full cooking videos, and multi-view images of the finished dishes, all paired with detailed nutritional annotations, as illustrated in Figure 2. Unlike existing datasets, OmniFood8K employs a unified, process-oriented design that preserves causal links from ingredient preparation to final presentation and provides comprehensive nutritional annotations. In addition, to further improve the model's nutritional prediction capability, we construct NutritionSynth-115K, a large-scale synthetic dataset with compositional diversity and precise nutritional annotations.

Second, we propose an end-to-end framework that predicts nutritional information directly from a single RGB image. Compared with depth-sensor-based approaches, our method is more generalizable and scalable for real-world dietary assessment. Specifically, we first employ a pre-trained depth estimation model [62] to predict the depth information of food. To correct scale bias and local structural errors, we design a Scale-Shift Residual Adapter (SSRA) for consistent global and local calibration. We then employ a Frequency-Aligned Fusion Module (FAFM) to hierarchically fuse RGB and adapted depth features, aligning multi-modal representations in the frequency domain for enhanced cross-modal learning. Finally, a Mask-based Prediction Head (MPH) leverages dynamic channel selection and region-aware attention to emphasize key ingredient regions, improving nutrition prediction accuracy.

Overall, our contributions are summarized as follows:

- We introduce OmniFood8K, a comprehensive multimodal food dataset with detailed nutritional annotations, covering ingredients, recipes, cooking videos, and multi-view food images. Additionally, we construct NutritionSynth-115K, a large-scale synthetic dataset featuring diverse food compositions and precise nutritional annotations.
- We propose an end-to-end framework for predicting food nutritional information from a single RGB image. It incorporates a Scale-Shift Residual Adapter (SSRA) to refine depth estimation and achieve consistent calibration of both global scale and local geometry.
- We further design a Frequency-Aligned Fusion Module (FAFM) to hierarchically fuse RGB and depth features in the frequency domain, and a Mask-based Prediction Head (MPH) that dynamically selects informative channels, thereby enhancing the accuracy of nutritional prediction.
- Extensive experiments on multiple datasets demonstrate that our method outperforms state-of-the-art approaches, validating its effectiveness and scalability for food nutrition assessment.

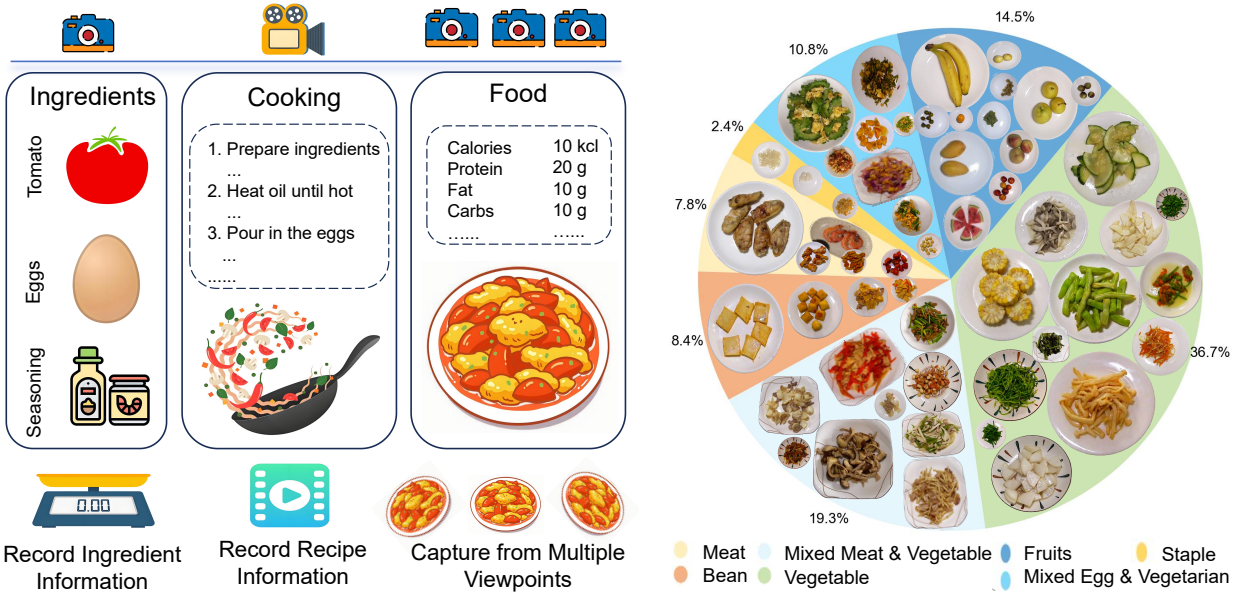
2. Related work

2.1. Food Datasets

Recent advances in food computing have led to the release of numerous publicly available datasets [1, 15], which have substantially promoted research across multiple domains, such as food recognition, cross-modal analysis, and nutritional estimation [4, 12, 40, 52]. Representative datasets such as Food-101 [5], VIREO Food-172 [6], ISIA Food-500 [30], and Food2K [31] cover diverse food categories and have been widely adopted for recognition and classification tasks [36, 67, 68]. Additionally, Recipe1M [41] aligns food images with textual recipes, serving as a fundamental resource for cross-modal retrieval and recipe generation tasks. Nutrition5k [53] includes detailed nutritional annotations and has been instrumental in advancing research on food nutrition assessment. However, its primary focus on Western dishes, with limited coverage of complex cuisines such as Chinese food, constrains its general applicability. FoodSeg103 [61] is a dataset designed for food image segmentation, aiming to facilitate fine-grained understanding of food components. MetaFood3D [7] is a multimodal dataset comprising 637 3D food objects across 108 categories, accompanied by detailed nutritional information and diverse visual modalities. The FastFood dataset [37] collects food images and their corresponding nutritional information from official fast-food brand websites, providing valuable data for nutrition analysis and related studies.

2.2. Food Nutrition Assessment

Traditional nutrition assessment methods are time-consuming, labor-intensive, and often limited in predictive accuracy [9, 39, 54]. With advances in computer vision, AI-based approaches for nutrition prediction have increasingly emerged [16, 21, 42, 46, 53, 56]. NR *et*



(a) Food Data Collection Process **(b) Distribution of Food Categories**

Figure 2. Overview of the OmniFood8K dataset: data collection process and category distribution.

al. [34] employed CNN to automatically recognize food and predict its nutritional information. Swin-Nutrition [42] introduced an efficient, non-destructive AI-based framework for accurately predicting multiple food nutrient components using Swin Transformer features. Vinod *et al.* [55] integrated energy density maps with depth information to effectively improve calorie estimation accuracy. Shao *et al.* [44] proposed estimating food energy through 3D shape reconstruction from a single-view image, achieving promising results. Feng *et al.* [14] proposed a method that incorporates ingredient information to complement visual features, thereby improving the accuracy of nutrition prediction and achieving promising results. RoDE [21] introduced a multi-expert framework to improve the accuracy and efficiency of large-scale food multi-modal models for tasks such as nutrition estimation. Ma [26] *et al.* proposed FBFPN, a network that fuses RGB and depth images via a bidirectional feature pyramid to improve nutrition estimation accuracy.

3. OmniFood8K Dataset Construction

3.1. Motivation

Existing nutrition datasets primarily focus on Western foods and lack sufficient representation of complex cuisines such as Chinese dishes, limiting their applicability to accurate nutritional estimation across diverse dietary contexts. To address these limitations, we introduce OmniFood8K, a comprehensive multimodal dataset for practical nutrition assessment, supporting diverse tasks such as nutrition estimation, food image generation, ingredient recognition, and recipe generation.

3.2. Data Collection

The dataset comprises 8,036 food samples, representing a wide range of commonly consumed Chinese dishes. Data were collected from real-world cooking scenarios, including raw ingredients, structured recipes, cooking videos, and final dish presentations. The data acquisition process, illustrated in Figure 2, follows a structured and systematic workflow. Raw ingredients were first weighed and photographed, followed by video recording of the cooking process and textual documentation of the recipes. Finally, the prepared food was captured simultaneously from six viewpoints at a horizontal height of 50 cm [49]. This height approximates the typical elbow level of an adult, making the setup convenient and practical for everyday use in real-life scenarios. Two cameras were positioned directly above the food, while the remaining four were placed at the front, back, left, and right sides.

3.3. Annotation

For each food scene, ingredients were first photographed and their corresponding weights recorded to enable precise computation of nutritional information. Based on the recorded ingredient weights, the nutritional contents of each scene, including calories, protein, fat, and carbohydrates, were calculated by referencing official nutritional databases to ensure accurate estimation of each food item's nutrient composition. Recipes were documented using structured

Table 1. Comparison between the proposed dataset and existing food datasets.

Dataset	Categories	Size	Ingredient Image	Food Image	Multi-view Food Image	Recipe	Cooking Video	Nutritional Info
Food101 [5]	101	101,000	-	✓	-	-	-	-
VIREO Food-172 [6]	172	110,241	-	✓	-	✓	-	-
Recipe1M [41]	1480	1,047,000	-	✓	-	✓	-	-
Yummly-66K [27]	10	66,615	-	✓	-	✓	-	-
ISIA Food-200 [29]	200	197,323	-	✓	-	-	-	-
ISIA Food-500 [30]	500	399,726	-	✓	-	-	-	-
Food2K [31]	2000	1,036,564	-	✓	-	-	-	-
YouCook2 [70]	89	2000	✓	✓	-	✓	✓	-
Menu-Match [2]	41	646	-	✓	-	-	-	✓
Fang *et al.* [13]	3	45	-	✓	-	-	-	✓
Nutrition5k [53]	-	5066	-	✓	✓	-	-	✓
MetaFood3D [7]	108	637	-	✓	✓	-	-	✓
OmniFood8K (**Ours**)	165	8,036	✓	✓	✓	✓	✓	✓
NutritionSynth-115K (**Ours**)	165	115,000	✓	✓	-	✓	✓	✓

Table 2. Data types and corresponding applications in the OmniFood8K dataset.

Data Type	**Applications**
Ingredient Images	Ingredient Recognition, Detection, Quantity Estimation.
Recipe Descriptions	Recipe Generation, Cross-Modal Retrieval, Cooking Reasoning.
Cooking Process Videos	Action Recognition, Cooking Stage Understanding, Multimodal Alignment with Recipes.
Multi-view Images of Food	Food Recognition, Detection, Image Synthesis, 3D-Aware Food Understanding.
Food Nutritional Data	Nutrition Prediction, Dietary Assessment, Health-oriented Analysis.

textual descriptions, while the entire cooking process was comprehensively recorded on video. Additionally, key steps in each recipe were captured as corresponding images to support multimodal analysis.

3.4. Dataset Characteristics

OmniFood8K provides nutritional annotations for 8,036 food samples, accompanied by diverse multimodal data, whose types and corresponding applications are summarized in Table 2. By spanning the entire food preparation pipeline, OmniFood8K serves as a comprehensive resource for research in food computing and nutritional analysis. Table 1 presents a comparison of OmniFood8K with existing food datasets across multiple characteristics. Notably, OmniFood8K is the first dataset to cover the entire food preparation process, from raw ingredients to finished food. Its main features are summarized as follows:

- **Multimodal**: OmniFood8K includes images, structured recipe descriptions, cooking videos, and detailed nutritional annotations.
- **Full Pipeline Coverage**: OmniFood8K spans the complete cooking process, from annotated ingredient images and weights to multi-view final food images, accompanied by recipe descriptions and cooking videos.
- **Causal Relationship**: OmniFood8K preserves the causal relationship from raw ingredients to finished dishes through recipe descriptions and cooking videos.
- **Accuracy:** Unlike other datasets [21, 37] that rely on web-scraped labels or LLM-generated annotations, OmniFood8K provides labels based on real-world weight measurements and standardized nutritional database calculations, ensuring higher accuracy.

3.5. NutritionSynth-115K

We performed foreground segmentation on OmniFood8K food images to isolate individual items. These segmented items were subsequently cropped or recombined according to their original proportions, allowing the synthesized samples to faithfully reflect the nutritional composition of each meal. Through this process, we constructed a large-scale and controllable synthetic dataset, named NutritionSynth-115K, which contains 115,000 images with precise nutritional annotations.

4. Method

The overall architecture of the proposed method is illustrated in Figure 3. A single RGB image is first fed into a depth estimation model to produce an initial depth map, which is subsequently refined using the proposed Scale-Shift Residual Adapter (SSRA) module for both global and

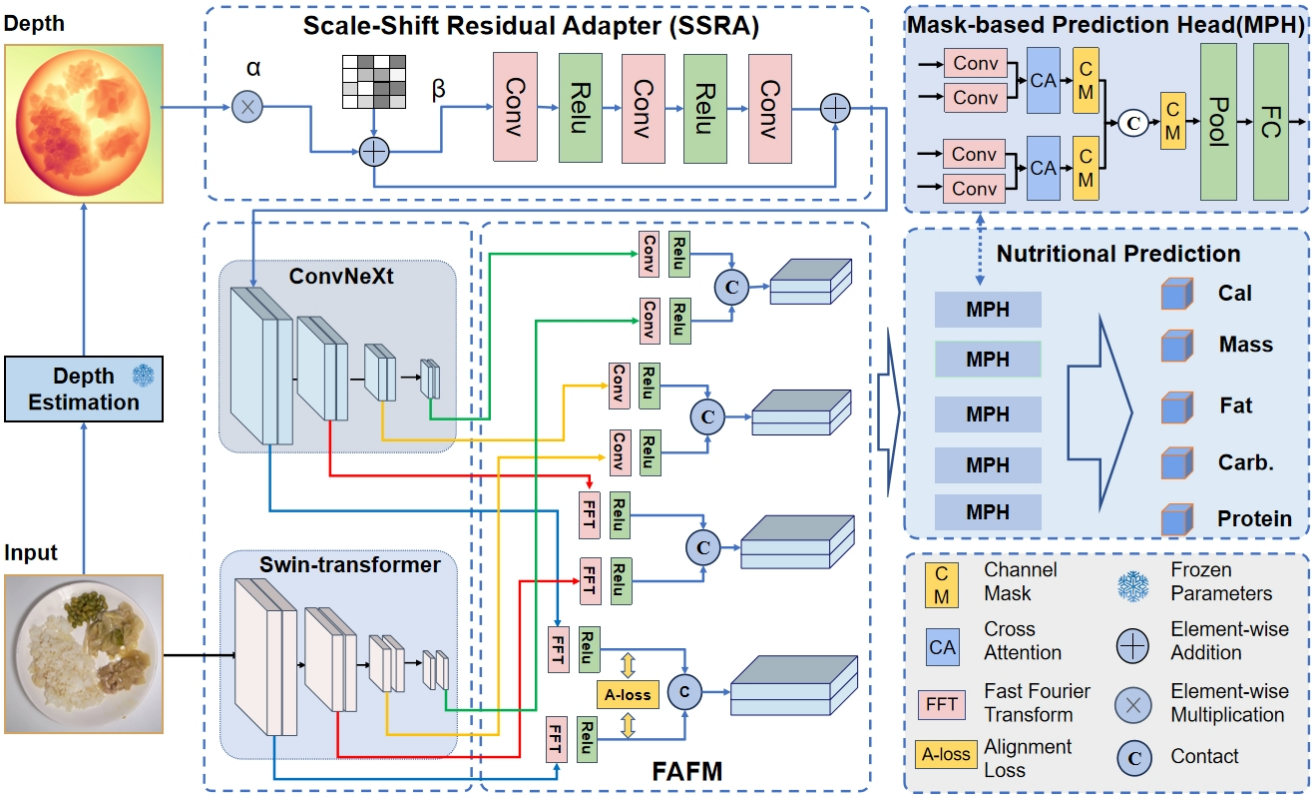


Figure 3. Overview of the proposed method. The figure illustrates the overall pipeline of our method, consisting of three proposed modules: the Scale-Shift Residual Adapter (SSRA), Frequency-Aligned Fusion Module (FAFM), and Mask-based Prediction Head (MPH).

local corrections. The RGB image and the refined depth map are then used to extract multi-scale features, which are fused by the Frequency-Aligned Fusion Module (FAFM) through frequency-domain alignment of multimodal representations. Finally, the Mask-based Prediction Head (MPH) predicts the nutritional information, forming an end-to-end framework for nutrition estimation.

4.1. Scale-Shift Residual Adapter (SSRA)

To ensure practical applicability in daily scenarios, we focus on nutrition estimation from a single RGB image. Since RGB images inherently lack explicit depth cues, we employ a monocular depth estimation model [62] to infer depth information. To further mitigate estimation errors, we design a lightweight Scale-Shift Residual Adapter (SSRA) that efficiently refines both global and local structures.

Specifically, at the global level, we introduce a learnable scaling factor α and shift parameter β to align the overall distribution of the predicted depth map d_{mono}:

$$d_{global} = \alpha \cdot d_{mono} + \beta. \tag{1}$$

This step effectively mitigates the common issues of scale ambiguity and offset errors in monocular depth prediction. Building upon this, a shallow convolutional residual refinement module $f_\theta(\cdot)$ is employed to enhance local structural details and correct geometric distortions. The refinement process can be expressed as:

$$d_{res} = f_\theta(d_{global}). \tag{2}$$

The final adapted depth is then obtained as:

$$d_{out} = d_{global} + d_{res}. \tag{3}$$

By combining global alignment with local refinement, the proposed SSRA ensures consistency in overall depth estimation while improving local precision, thus enhancing the accuracy of nutrition estimation tasks.

4.2. Frequency-Aligned Fusion Module (FAFM)

To effectively integrate complementary RGB and depth information while preserving feature consistency, we propose a Frequency-Aligned Fusion Module (FAFM). This module fuses multi-frequency features and enforces cross-modal consistency via an inter-modal alignment loss.

Given RGB and depth feature maps $r, d \in \mathbb{R}^{C\times H\times W}$, the module first applies a 2D Fast Fourier Transform (FFT) to obtain the frequency representations $R_f = \mathcal{F}(r)$ and

$D_f = \mathcal{F}(d)$. Based on a predefined frequency threshold κ, a low-frequency mask M_L is constructed to separate low-frequency (global structure) and high-frequency (local detail) components:

$$R_L = \mathcal{F}^{-1}(R_f \odot M_L), \quad R_H = \mathcal{F}^{-1}(R_f \odot (1 - M_L)), \tag{4}$$

$$D_L = \mathcal{F}^{-1}(D_f \odot M_L), \quad D_H = \mathcal{F}^{-1}(D_f \odot (1 - M_L)). \tag{5}$$

The high-frequency and low-frequency components from both modalities are then fused via element-wise addition:

$$F_H = R_H + D_H, \quad F_L = R_L + D_L. \tag{6}$$

The fused high-frequency and low-frequency features are concatenated along the channel dimension and passed through a learnable convolutional layer to adaptively compress and integrate them, producing the final cross-modal feature representation:

$$O = \text{Conv}([F_H, F_L]). \tag{7}$$

In this design, high-frequency components enhance texture and local details, while low-frequency components preserve global structure, achieving Multi-Frequency Fusion.

To enhance semantic consistency between RGB and depth features, we introduce an inter-modal alignment loss $\mathcal{L}_{align}$, defined as:

$$\mathcal{L}_{align} = -\frac{1}{N} \sum_{i=1}^{N} \log \frac{\exp(\text{sim}(\mathbf{f}_i^r, \mathbf{f}_i^d)/\tau)}{\sum_{j=1}^{N} \exp(\text{sim}(\mathbf{f}_i^r, \mathbf{f}_j^d)/\tau)}, \tag{8}$$

where $\mathbf{f}_i^r$ and $\mathbf{f}_i^d$ are the global feature vectors of the i-th sample in RGB and depth modalities, $\text{sim}(\cdot, \cdot)$ denotes cosine similarity, N is the batch size, and τ is a temperature parameter. This loss encourages features from the same sample across modalities to be close, while separating features from different samples, thus improving cross-modal representation for subsequent fusion.

4.3. Mask-based Prediction Head (MPH)

To fully leverage multi-scale features and emphasize key ingredient regions, we propose the Mask-based Prediction Head (MPH). This module takes feature maps from different sources as input, including multi-scale RGB features and semantic features. First, MPH applies convolutional operations and adaptive average pooling to each feature map to extract high-dimensional representations and unify the spatial dimensions to a fixed size, ensuring the controllability and stability of subsequent fusion operations. Next, MPH employs a Cross-Attention mechanism to enable information interaction between RGB and semantic features, allowing the model to capture correlations across different modalities. This is followed by Gated Fusion, which further reweights the fused features to highlight the most critical channels. To further enhance the focus on key ingredient regions, MPH introduces a dynamic channel mask (Channel Mask), which automatically selects the most informative channels, thereby preserving the features most relevant to the target nutritional information during fusion. Finally, the RGB and semantic features are integrated through a global fusion module and processed by global pooling and a fully connected layer to produce the predicted nutritional values.

4.4. Loss Function

In the nutrient prediction task, we adopt a Dynamic Task-weighted Loss [63] for nutrient estimation, which dynamically adjusts task weights according to the current prediction difficulty. The nutrient prediction loss is formulated as:

$$\mathcal{L}_{\text{nutri}} = \sum_{i=1}^{n} w_i \cdot \text{PMAE}(\text{nutri}[i]), \tag{9}$$

where n is the number of nutrient components, w_i is the dynamically updated weight for the i-th nutrient task, and PMAE denotes the Percentage Mean Absolute Error. Specifically, each task i corresponds to a nutrient component such as Calorie, Mass, Fat, Carbohydrate, or Protein. The dynamic weights are updated according to the task difficulty via a KPI metric:

$$\begin{aligned} \text{KPI}_i &= \frac{1}{1 - \text{PMAE}(\text{nutri}[i])}, \\ w_i^{(t)} &= \alpha \cdot \text{KPI}_i^{(t)} + (1 - \alpha) \cdot w_i^{(t-1)}, \end{aligned} \tag{10}$$

where t denotes the training iteration, and α is a smoothing factor controlling the update momentum, which is empirically set to 0.3 in our experiments.

To enhance cross-modal representation consistency, we further introduce an inter-modal alignment loss $\mathcal{L}_{\text{align}}$ (Eq. 8), applied to fused RGB–Depth features. Finally, the total loss for training the nutrient prediction model is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{nutri}} + \lambda \, \mathcal{L}_{\text{align}}, \tag{11}$$

where λ balances the contribution of the inter-modal alignment loss. This formulation enables joint optimization of nutrient prediction and cross-modal feature alignment, thereby improving the model's predictive performance in multimodal fusion.

5. Experiments

5.1. Experimental Setup

1) Evaluation Metrics. The percentage of mean absolute error (PMAE) is adopted to evaluate the prediction performance of our method. This metric measures the relative

Table 3. Comparison of PMAE (%) values of different baselines on the OmniFood8K dataset.

Input	Method	Calories	Mass	Fat	Carb.	Protein	Mean
RGB images	Inception V3 [50]	17.2	10.0	17.8	13.8	20.5	16.0
	ResNet-50 [17]	10.2	5.1	11.0	11.1	14.8	10.4
	ResNet-101 [17]	10.9	6.0	12.2	9.1	15.2	10.6
	DenseNet-121 [19]	12.1	7.4	13.7	10.5	18.2	12.4
	DenseNet-161 [19]	12.2	7.6	12.7	12.6	16.3	12.3
	EfficientNet [51]	27.1	10.4	28.3	38.9	40.0	28.9
	MobileNetV3 [18]	17.1	8.5	18.5	21.2	27.1	18.5
	Swin-Transformer [25]	27.7	15.1	29.6	42.6	41.0	31.2
	ConvNeXt [24]	10.1	5.6	10.8	9.5	13.3	9.8
	ConvNeXt V2 [60]	15.5	6.4	11.3	9.3	**12.6**	11.2
	Ours	**8.8**	**4.5**	**9.5**	**8.0**	**12.6**	**8.6**

Table 4. Comparison of PMAE (%) values of different baselines on the Nutrition5k dataset.

Input	Method	Calories	Mass	Fat	Carb.	Protein	Mean
RGB images	Google-Nutrition-rgb [53]	26.1	18.8	34.2	31.9	29.5	29.1
	Coarse-to-Fine Nutrition [56]	24.1	19.4	36.0	32.1	33.5	29.0
	Swin-Nutrition [42]	16.2	13.7	24.9	21.8	25.4	20.4
	Portion-Nutrition [44]	15.8	-	-	-	-	-
	RoDE [21]	52.4	38.4	67.1	47.8	53.9	51.9
	DPF-Nutrition [16]	14.7	10.6	22.6	20.7	20.2	17.8
RGB-D images	CMX [65]	21.8	20.7	34.8	37.0	33.2	29.5
	HINet [3]	24.5	25.2	43.4	39.9	38.8	34.3
	CDINet [64]	21.1	20.4	37.1	37.1	32.8	29.7
	DEFNet [71]	32.7	34.2	48.9	40.3	43.8	39.9
	TriTransNet [23]	22.1	20.1	37.5	34.8	38.0	30.5
	Deliver [66]	29.5	25.9	48.3	47.7	46.1	39.5
	Google-Nutrition-rgbd [53]	18.8	18.9	**18.1**	23.8	20.9	20.1
	Domain Adaptation-Nutrition [55]	16.8	-	-	-	-	-
	RGB-D Net [43]	15.0	10.8	23.5	22.4	21.0	18.5
	IMIR-Net [33]	14.5	10.4	21.8	20.4	20.0	17.4
	FBFPN [26]	**14.0**	10.3	22.6	19.5	20.2	17.3
RGB images	**Ours**	14.1	**10.2**	21.0	**18.9**	**18.4**	**16.5**

deviation between predicted and true values, with smaller values indicating more accurate predictions. PMAE is defined as:

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_t - y_p|, \tag{12}$$

$$\text{PMAE} = \frac{\text{MAE}}{\frac{1}{N}\sum_{i=1}^{N} y_t} \times 100\%, \tag{13}$$

where y_t denotes the true value, y_p denotes the predicted value, and N denotes the total number of samples. A smaller value indicates higher prediction accuracy and better model performance.

2) Baseline Methods. To evaluate the effectiveness of our proposed method, we conduct a comparison with recent state-of-the-art approaches. On the OmniFood8K dataset, we select representative models including [17–19, 25, 50, 51]. For the Nutrition5k dataset, we compared our method with recent approaches, including Google-Nutrition [53], Portion-Nutrition [44], Coarse-to-Fine Nutrition [56], Swin-Nutrition [42], DPF-Nutrition [16], RGB-D Net [43], Domain Adaptation-Nutrition [55], IMIR-Net [33], FBFPN [26]. We also compare our method with prior image fusion approaches [3, 23, 64–66, 71] .

Table 5. Ablation study of the proposed modules on the Nutrition5k dataset.

Baseline	FAFM	SSRA	MPH	Calories	Mass	Fat	Carb.	Protein	Mean
✓				14.8	11.3	23.6	20.1	19.7	17.9
✓	✓			14.4	**10.2**	21.7	19.7	19.2	17.0
✓	✓	✓		14.2	10.4	21.9	19.2	18.6	16.8
✓	✓	✓	✓	**14.1**	**10.2**	**21.0**	**18.9**	**18.4**	**16.5**

3) Implementation Details. All models were implemented in PyTorch and trained on an NVIDIA A100 GPU. The Adam optimizer was used with a weight decay of 1e-5 to prevent overfitting. A cosine annealing scheduler adjusted the learning rate during training. Models were trained for 150 epochs with a batch size of 8. The OmniFood8K and Nutrition5k datasets are split into training and testing sets at ratios of 7:3 and 5:1 [16, 43], respectively.

5.2. Experimental Results

To validate the effectiveness of the proposed method, we conduct experiments on multiple food nutrition estimation datasets. Table 3 presents the comparison with different baseline methods on our proposed OmniFood8K dataset. It can be observed that our method achieves the best performance across multiple metrics, with the lowest PMAE, indicating superior prediction accuracy. Table 4 reports the comparison results on the Nutrition5k dataset. Our method outperforms existing state-of-the-art approaches on the PMAE metric across multiple nutrient components, achieving the best overall performance. Notably, even when compared with a number of state-of-the-art methods that directly leverage RGB-D inputs, our method still achieves superior performance, further demonstrating its effectiveness and robustness.

As shown in Tables 3 and 4, our method achieves a lower PMAE on OmniFood8K than on Nutrition5k, demonstrating superior predictive accuracy. This improvement can be attributed to the larger scale of OmniFood8K, which contains 8,036 images (5,600 used for training) compared to Nutrition5k's 3,500 images (2,800 for training) [14, 43]. The roughly twofold increase in training samples exposes the model to more diverse food appearances and nutritional compositions, enhancing feature learning and prediction performance.

5.3. Ablation Study

To evaluate the contribution of each component in our method, we conduct an ablation study on the Nutrition5k dataset. We gradually add the proposed components and observe the resulting performance changes. As shown in Table 5, the model's performance steadily improves with the inclusion of each component, demonstrating the effectiveness of our method.

Table 6. Evaluation of our method pre-trained on NutritionSynth-115K and fine-tuned on OmniFood8K.

Method	Calories	Mass	Fat	Carb.	Protein	Mean
Ours	8.8	**4.5**	9.5	8.0	12.6	8.6
+ Pre-trained	**8.0**	**4.5**	**9.0**	**7.4**	**10.0**	**7.8**

5.4. Pretraining on NutritionSynth-115K

To enhance the performance of our method on the OmniFood8K dataset, it is first pretrained on NutritionSynth-115K. This pretraining enables the model to capture rich food features and multimodal representations, providing a robust initialization for subsequent fine-tuning on OmniFood8K. As shown in Table 6, pretraining significantly improves predictive performance, demonstrating its effectiveness in enhancing generalization.

6. Conclusion

To enable practical and accessible nutritional estimation, we presented an end-to-end framework capable of predicting nutritional information directly from a single RGB image. In addition, we introduced OmniFood8K, a comprehensive dataset containing 8,036 real-world food samples with multi-view images, ingredient weights, recipes, cooking videos, and detailed nutritional annotations. To further improve nutritional prediction, we also introduced NutritionSynth-115K, a large-scale synthetic dataset that preserves accurate nutritional labels. Our framework first predicts monocular depth, which is globally and locally refined by the Scale-Shift Residual Adapter (SSRA) to ensure scale consistency and preserve structural accuracy. The Frequency-Aligned Fusion Module (FAFM) hierarchically fuses RGB and depth features in the frequency domain, capturing both spatial and cross-modal semantic information. Finally, the Mask-based Prediction Head (MPH) dynamically selects informative channels to improve prediction accuracy. Extensive experiments across multiple datasets demonstrate that our approach consistently outperforms existing methods in nutritional prediction, validating its effectiveness and generalizability.

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